

Chapter 21 - The Paula DMA Engine

Paula is the four-channel sample DMA engine. It reads signed 8-bit sample bytes from memory and feeds them to four SoundChip DAC channels at rates chosen by Paula-style period values. The program supplies the sample address, length, period, and volume, then arms DMA with one control write.

Use MOD playback in Chapter 19 for tracker modules. Use WAV playback in Chapter 20 for RIFF/WAVE files. Use Paula when you want exact control of raw sample buffers, completion status, double-buffering, and M68K level 3 interrupts.

21.1 First sound

Type this program. It builds four signed 8-bit sine samples in memory and plays them as a short chord through all four Paula channels.

```
10 REM PAULA FOUR CHANNEL CHORD
20 POKE &H000F0800,1
30 A=&H00120000:N=4096:P=253
40 REM BUILD FOUR SIGNED 8 BIT SAMPLES
50 FOR C=0 TO 3
60 F=220
70 IF C=1 THEN F=277
80 IF C=2 THEN F=330
90 IF C=3 THEN F=440
100 FOR I=0 TO N-1
110 V=INT(SIN(I*TWOPI*F/14019)*100)
120 IF V<0 THEN V=V+256
130 POKE8 A+C*N+I,V
140 NEXT I
150 NEXT C
160 REM PTR, LEN, PERIOD, VOLUME
170 FOR C=0 TO 3
180 B=&H000F2260+C*16
190 POKE B,A+C*N
200 POKE B+4,N/2
210 POKE B+8,P
220 POKE B+12,40
230 NEXT C
240 REM CLEAR STATUS, THEN ARM ALL CHANNELS
250 POKE &H000F22A4,15
260 POKE &H000F22A0,&H800F
270 FOR T=1 TO 3000
280 NEXT T
290 PRINT PEEK(&H000F22A4)
```

You should hear a brief four-note chord. Line 290 prints the completion status; after the chord has ended, bits 0-3 are set.

Lines 40 to 150 build four separate sample buffers. Lines 160 to 230 write each channel's pointer, word length, period, and volume. Line 250 clears old completion bits, and line 260 arms all four DMA channels at once.

21.2 What Paula can produce

Item	Value
Channels	4
Sample format	Signed 8-bit PCM
Length unit	Words, where one word is two sample bytes
Period clock	3546895 / period samples per second
Volume	0 - 64 per channel
Channel output	Paula channel n feeds SoundChip DAC channel n
Completion status	One bit per channel
Interrupt	Optional M68K level 3 interrupt on completion

Sample bytes are signed. \$00 is silence, \$7F is a large positive sample, and \$80 is a large negative sample. In BASIC, add 256 before POKE8 when a calculated sample is negative.

21.3 Register block

The Paula register block is \$F2260-\$F22AF. Channels are 16 bytes apart.

Channel	Base	End
0	\$F2260	\$F226F
1	\$F2270	\$F227F
2	\$F2280	\$F228F
3	\$F2290	\$F229F

Each channel has the same layout:

Offset	Name	Access	Purpose
\$00	PTR	write/read	Sample start address. Odd addresses are masked down to even.
\$04	LEN	write/read	Length in words, so bytes divided by two.
\$08	PER	write/read	Paula period. Output rate is 3546895 / PER.
\$0C	VOL	write/read	Volume 0 - 64; larger writes clamp to 64.

The global registers are:

Address	Name	Access	Purpose
\$F22A0	AROS_AUD_DMACON	write/read	Set or clear active DMA channel bits.
\$F22A4	AROS_AUD_STATUS	write/read	Completion and error-style channel flags.
\$F22A8	AROS_AUD_INTENA	write/read	Set or clear completion interrupt enables.

The AROS_AUD_* names are the canonical register names for this Paula block.

21.4 DMACON and INTENA

AROS_AUD_DMACON and AROS_AUD_INTENA use the same set-or-clear format.

Bit	Meaning
15	1 means set the selected bits; 0 means clear them.
0	Channel 0 mask.
1	Channel 1 mask.
2	Channel 2 mask.
3	Channel 3 mask.

Examples:

```
10 POKE &H000F22A0,&H8001:REM ENABLE CHANNEL 0
20 POKE &H000F22A0,&H800F:REM ENABLE CHANNELS 0-3
30 POKE &H000F22A0,1:REM DISABLE CHANNEL 0
40 POKE &H000F22A8,&H8001:REM ENABLE CH0 COMPLETION IRQ
50 POKE &H000F22A8,1:REM DISABLE CH0 COMPLETION IRQ
```

When a channel reaches the end of its buffer, the matching bit in AROS_AUD_STATUS is set. Clear status bits by writing 1 bits back:

```
10 REM PAULA STATUS CLEAR
20 PRINT PEEK(&H000F22A4)
30 POKE &H000F22A4,15
40 PRINT PEEK(&H000F22A4)
```

The status register is cleared by writing 1 bits back to it. Line 30 clears all four channel bits without changing DMACON.

If AROS_AUD_INTENA has the matching channel bit set, a completed buffer also asserts M68K interrupt level 3.

21.5 Setup order

To start one channel from a clean state:

1. Enable audio with `POKE &H000F0800,1`.
2. Write signed sample bytes to memory.
3. Write PTR, LEN, PER, and VOL.
4. Clear any old status bit by writing to AROS_AUD_STATUS.
5. Write \$8000 plus the channel mask to AROS_AUD_DMACON.

This is channel 0 only:

```

10 REM PAULA CHANNEL 0 SETUP
20 A=&H00124000:N=1024
30 REM BUILD A SIGNED SAMPLE BUFFER
40 FOR I=0 TO N-1
50 V=INT(SIN(I*TWOPI*330/14019)*100)
60 IF V<0 THEN V=V+256
70 POKE8 A+I,V
80 NEXT I
90 REM PTR, LEN, PERIOD, VOLUME
100 POKE &H000F2260,A
110 POKE &H000F2264,N/2
120 POKE &H000F2268,253
130 POKE &H000F226C,64
140 REM CLEAR OLD STATUS AND ARM CH0
150 POKE &H000F22A4,1
160 POKE &H000F22A0,&H8001

```

If LEN is zero, or PER is still zero when the channel is armed, the channel is not accepted and its status bit is set. A direct write of zero to PER is ignored, so an existing non-zero period is not erased by accident.

21.6 Period and pitch

The sample rate is:

$$\text{rate} = 3546895 / \text{PER}$$

Useful periods:

Period	Approximate rate	Use
124	28604 Hz	Bright samples.
161	22030 Hz	Common sampled effects.
253	14019 Hz	Classic tracker note rate.
508	6982 Hz	Low-rate speech or drums.
1015	3494 Hz	Very low-rate speech.

The fetch engine runs from the Paula clock and writes to the mixer at the machine audio rate. Fractional phase is preserved across a staged next buffer, which keeps double-buffered streams smooth.

21.7 Latching and double-buffering

When AROS_AUD_DMACON arms a channel, Paula latches the current PTR, LEN, PER, and VOL as the active buffer. Writes to those registers while the channel is active do not disturb the current buffer. They stage the next buffer instead.

When the active buffer ends:

- The channel status bit is set.
- If a staged next buffer has non-zero length and period, Paula switches to it without clearing the channel's DMACON bit.
- If there is no valid next buffer, the channel stops and its DMACON bit clears.

This fragment stages a second channel-0 buffer while the first is playing:

```
10 REM FIRST BUFFER IS ALREADY ARMED ON CH0
20 REM STAGE NEXT PTR, LEN, PERIOD, VOLUME
30 POKE &H000F2260,&H00126000
40 POKE &H000F2264,512
50 POKE &H000F2268,253
60 POKE &H000F226C,48
```

The new values take effect only after the current buffer is exhausted. That is the normal way to stream audio without a gap.

21.8 Stopping and error cases

To stop channel 0:

```
10 POKE &H000F22A0,1
```

To stop all four channels:

```
10 POKE &H000F22A0,15
```

If the sample pointer is beyond readable memory, Paula mutes the channel, stops it, clears its DMA CON bit, and sets the status bit. If the matching interrupt-enable bit is set, this also raises the level 3 interrupt.

Pointers are even-aligned by masking off bit 0. Length is not rounded: LEN=5 means ten sample bytes. Keep buffers even-sized unless you are deliberately using an odd word count.

21.9 Limits

Paula plays raw signed 8-bit samples only. It does not parse WAV headers, MOD patterns, or sample-loop metadata. Use the WAV and MOD chapters for those formats.

Volume changes written during playback are staged with the next buffer, because the active buffer uses its latched volume. For an immediate volume effect, stop and re-arm the channel or use a SoundChip DAC path from Chapter 12.